



Statistical evaluation of using the new generation of wind turbines in South Africa

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ABSTRACT

In the view of the latest status and the potential of developing wind energy in South Africa, the present study aims to perform technical–economic–environmental analysis on a wind turbine system with HOMER software using the 20-years average data of the wind speed obtained from NASA's database, for providing the electricity to residential buildings. The results showed that the Port Elizabeth station, had the lowest levelized cost of electricity (LCOE) with the value of -0.363 \$/kWh when using the EOLO wind turbine, and the Bloemfontein station had the highest LCOE with the value of 1.601 \$/kWh when using the Turby wind turbine. The results from the step-wise assessment ratio analysis (SWARA) weighting method demonstrated the wind penetration, total production, and capital cost as the most important sub-indices with the weights of 0.106, 0.095, and 0.091, respectively. Using the additive ratio assessment (ARAS), weighted sum method (WSM), and weighted aggregated sum product assessment (WASPAS) techniques, the cities under study were ranked, and the cities of East London and Bloemfontein were identified as the most suitable and the most unsuitable stations for the use of household-scale wind turbines, respectively.

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1. Introduction

Wind energy is a free, inexhaustible, non-pollutant renewable energy source (Madvar et al., 2019) that is used today to generate electricity (Alamdari et al., 2012; Shaban et al., 2020). When using wind energy on a large-scale, large areas of land, as well as big noisy turbines, are required (Riahi Zaniani et al., 2015; Jahangiri

et al., 2012). Therefore, nowadays, there is a great tendency toward the use of wind energy on a household scale.

Challenges such as exploitation of renewable energies in the long term, deciding on whether or not to expand the grid, studying the impact of seasonal changes, etc., are among the most important issues faced by the energy policy makers in South Africa, in order to be able to manage the energy crisis of the country. Despite the importance of these topics, a review regarding the use of wind energy on a household scale connected to the national electricity grid in South Africa reveals that very few studies have been performed so far.

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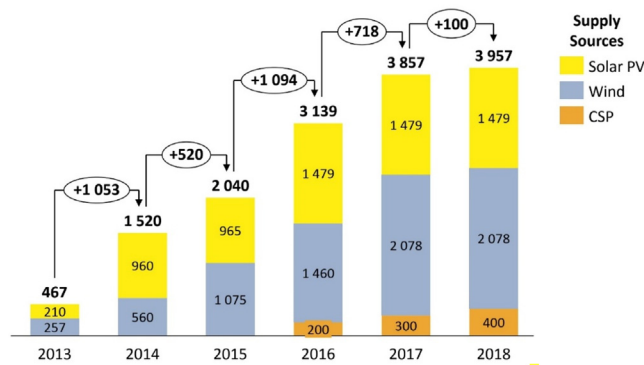


Fig. 1. Operational capacity (MW) of RE sources in SA (Calitz and Wright, 2019).

South Africa holds good wind potential, which is especially higher in coastal areas of the country. The average wind speed is equal to 6 m/s at an altitude of 10 m, and the whole country has potential annual wind energy of 60 TWh (Sichilalu et al., 2017).

According to Fig. 1, the total capacity of wind farms established in South Africa is equal to 2078 MW (Calitz and Wright, 2019), with an additional 840.9 MW under construction, with the largest wind farm in South Africa with the capacity of 148 MW under construction at Nxuba (Anon., 2020a). As can be seen from this Figure, the installed wind power capacity in South Africa is about 40% higher than its solar power. Another issue reflected in Fig. 1, is that the use of wind and solar powers has increased about 8.5 times, from 467 MW to 3957 MW, in the past six years. Of course, it is a negative point that during 2018, the growth of installed capacity of renewable electricity has almost stagnated. It is important because South Africa satisfies most of its needs using coal, despite having the third-largest gross domestic product (GDP).

The wind energy atlas of South Africa at an altitude of 100 m is presented in Fig. 2 (Anon., 2020b). As can be seen from this Figure, the southern one-third of the country is more suitable than its other regions for the use of wind energy. It is predicted that by the year 2030, about 11800 MW up to 12500 MW of the energy in South Africa will be generated using wind, which makes it necessary to investigate the potential of this energy at a household scale (Mukonza and Nhamo, 2018).

As seen in Fig. 3, the tariffs for wind and solar powers in South Africa, are at least 40% cheaper, compared to the fossil-fuel-based electricity (Anon., 2019).

South Africa aims to produce 18.2 GW of electricity using renewable resources by the year 2030. It includes 8.4 GW of solar electricity, 8.4 GW of wind electricity, 1 GW of concentrated solar energy, and 0.4 GW from other renewable sources (Aliyu et al., 2018). It means the number of people with no access to the electricity in the country will be zero.

The end consumers of electricity in South Africa are shown in Table 1. As can be seen from this Table, household electricity, with a share of 17.2% consumption of the total electricity generated in the country, ranks second among the sectors with the highest consumption, after the industrial electricity sector with a share of 37.7%.

The following are some of the similar researches previously performed in South Africa. All studies have been done recently, include techno-econo-enviro analysis and have used renewable energy to meet the needs.

In 2018, Madziga et al. (2018) presented a comparison, using the HOMER software, of three systems disconnected from the grid, namely the battery-solar cell (scenario 1), solar cell-diesel generator (scenario 2), and solar cell-diesel generator-battery

Table 1

South African end users electricity consumption by sectors (Du Plessis, 2015).

Sector	Percentage (%)
Domestic	17.2
Agriculture	2.6
Mining	15
Industries	37.7
Commerce	12.6
Transport	2.6
Others	12.3

(scenario 3), in order to bring electricity into a village in South Africa. The results of their research showed that with a total net present cost (NPC) equal to 140970.6 \$, the third scenario is not economical. The cost per kWh of the generated energy for the first, second, and third scenarios, were equal to 0.87 \$, 3.34 \$, and 4.55 \$, respectively. Moreover, with the annual emission of 6928 kg of CO₂ pollutant, the second scenario that does not use batteries was proved to be the worst scenario environmentally.

In 2018, Patel (2018) investigated bringing electricity to 3 villages in South Africa that would become very expensive to connect to the grid. One of these villages had a strong solar radiation potential, the other had a very strong wind potential, and the last one contained a strong hydro potential. The results of the study using the HOMER software demonstrated that the systems based on wind energy are not suitable for the electricity generation in small scales and that the hydro energy is the cheapest option for each village, but it brings social and environmental consequences due to its need for construction of dams, and thus the solar system was chosen as the most suitable option, due to its easier and faster implementation procedure. The cheapest solar, wind and micro-hydro systems had an LCOE per kWh, equal to 0.3715 \$, 0.4172 \$, and 0.1823 \$, respectively.

In 2018, Meje et al. (2018), designed, implemented, and controlled a hybrid system containing pico hydro, solar cells, wind turbine, and diesel generator, for small loads in the villages of South Africa. The energy sources were chosen based on the existing potential of renewable energy resources in the region. No simulations were performed in their study, and merely the components of the system were chosen.

In 2019, Ayodele and Munda (2019), evaluated the economic potential and the feasibility of producing hydrogen from the electrolysis of water and generating electricity using wind energy in South Africa. Data regarding the wind speed at an altitude of 60 m at 15 different stations were used to investigate the power generated using 11 different wind turbines. The results showed that the price range for each kilogram of hydrogen and one kWh of electricity are in the range of 1.4–39.55 \$, and 0.23–0.42 \$, respectively.

In 2019, Gabra et al. (2019), performed an economic-technical analysis on the wind micro-grid disconnected from the grid, in comparison with the solar cell and diesel generator, in the African continent. Their results showed that the use of wind energy in the Horn of Africa, and small regions in the southern parts of South Africa and the northwest of Africa, is cost-efficient. This cost efficiency holds for the use of diesel generators in the north and 2 vast regions in the west of Africa, while in other regions of Africa (the vast majority of the continent), the use of solar cells is more cost-efficient.

In 2019, Adefarati and Obikoya (2019), evaluated the potential of wind sources at the altitude of 50 m in 10 stations located in South Africa, and using the HOMER software, they also performed an economical analysis. Their results showed that the sites surveyed were in the range of the IV class in the international electrotechnical commission (IEC) wind classification. Moreover,

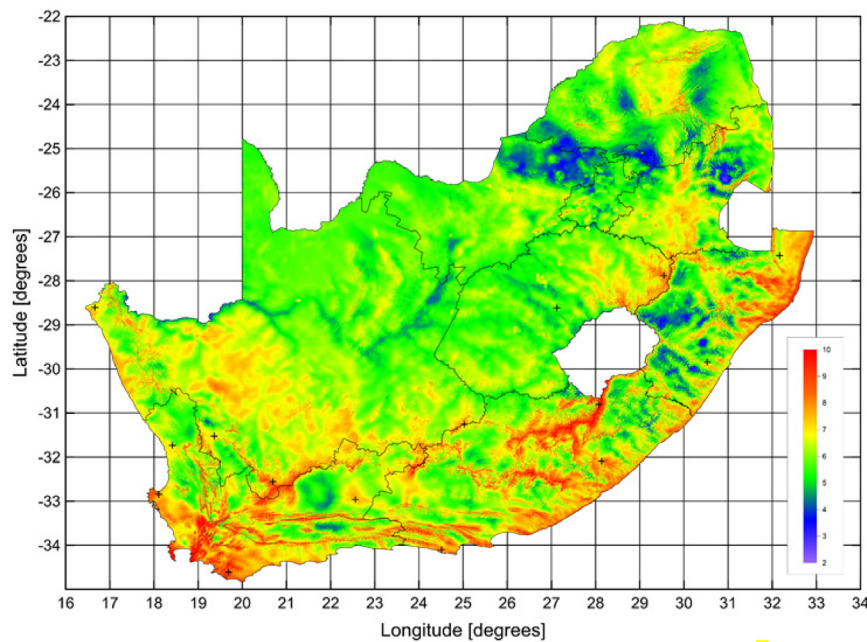


Fig. 2. South African Wind Atlas (Mean wind speed m/s @100 m) (Anon., 2020b).

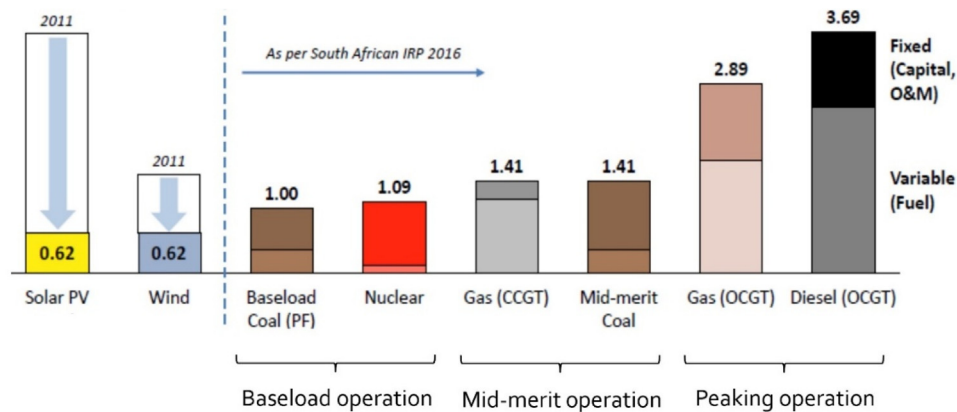


Fig. 3. Lifetime cost per energy unit in R/kWh (Anon., 2019).

the results showed that Cape Town, with an average annual wind speed of 6.39 m/s was the most suitable site.

In 2019, Babatunde et al. (2019) investigated the impact of temperature, direction, and solar tracker on electrification in South Africa at the household scale, using the HOMER software. The results demonstrated that the system without a tracker was more viable economically. Furthermore, in cases of considering and not considering the temperature effects, the dual axial tracker system, and the continuous horizontal axis tracker system, had the most annual electricity generation, respectively.

The main definition of decision making can be explained by the selection of a particular item from a set, so that the needs of decision makers are properly answered (Davarpanah et al., 2019). Today, decision-making methods such as WASPAS, ARAS and WSM are used to find optimal solutions to various problems. The following are some recent works in the field of renewable energies that have been done using these decision-making methods.

In 2017, Nie et al. (2017) used WASPAS method to select the optimal location of the wind-solar power plant due to the existence of many influential parameters as well as the existence of several locations for the project. The parameters studied were

natural resources, economic, traffic, environmental and social. The results showed that WASPAS method is well in line with reality and has been able to select the location of the power plant well.

In 2018, Rezaei et al. (2018) ranked the wind-solar hybrid power plant using the FTOPSIS method. Their study locations were 7 stations in Fars province in Iran. The four main criteria studied were economic status, social, geological and natural disaster, each of which had sub-criteria. WSM, AHP and DEA methods were used to confirm the results. Based on the results, Eghlid was selected as the most suitable city.

In 2020, Ghenai et al. (2020) examined the sustainability indicators of various renewable energy systems using a multi-criteria decision model and SWARA/ARAS method. The criteria evaluated were resources, environment, economic, social and technology, which had a total of 14 sub-criteria. The final results of the rankings showed that wind energy, solid oxide fuel cell, phosphoric acid fuel cell and solar energy are ranked first to fourth, respectively. Finally, they stated that the multi-criteria decision model, along with SWARA/ARAS method, is useful for evaluating the sustainability and sustainable development of renewable energies.

According to the aforementioned studies, it is clear that no wind system connected to the grid in South Africa has been investigated to date. Also, all of the wind turbines employed were conventional (with horizontal axis), which required a huge amount of expenditure to be installed in heights. No similar work was found regarding the ranking of stations based on technical and economic indices. Thus, in this work, using the experimental data obtained from four vertical-axis wind turbines and a tree-shaped wind turbine, and using the HOMER software, the power supply of 8 selected stations in four different regions of power transmission in South Africa, has been investigated for the first time. The use of real electricity tariffs in 2 seasons of high demand and low demand, and at during peak, standard and off peak times, as well as considering the penalties for pollutants caused by the national electricity grid, are among the other contributions of the present study, compared to the previous works in the literature. The weighting of the parameters using the SWARA method, and ranking the stations using ARAS, WSM, and WASPAS techniques were also performed. The main goal of the current study is trying to raise awareness among the energy decision-makers in South Africa so that they can make more informed decisions in regards to sustainable development and energy security of this country or in developing countries with similar energy development factors.

2. Required data

Given that one aim of this study is to provide a comparison between the per kWh price of the generated energy in the quadruple electrification zones in South Africa, as seen in Fig. 4, two stations are chosen within each zone in the study. Witbank and Klerksdorp stations are located in zone 1, Durban, and Bloemfontein stations are in zone 2, East London, and Port Elizabeth stations are in zone 3. Finally, Cape Town and George stations are located in zone 4. These stations are chosen because each of them is stations with economic, tourist, cultural, and so on, importance.

According to Fig. 5 (Anon., 2020c), and based on the standard price program for the tariffs approved by the South African Power Supply Commission (ESKOM), the prices of electrification are in two forms for different seasons. From September to May are considered the low demand months, while from June to August are considered the high demand months of the year due to an increase in the need for heating due to the winter season. As seen in Fig. 5, there exist three times of peak, normal, and off-peak for the electricity consumption during the weekdays, with Saturday and Sunday being different from the others. Moreover, Table 1 Appendix provides the price of electricity depending on the season, day of the week, and time of the day.

The assumptions that are input to the HOMER software, are given in Table 2 Appendix (Anon., 2020d; Jahangiri et al., 2017; Ebrahimi et al., 2019; Anon., 2020e; Pahlavan et al., 2019). Due to the fact that the wind speed data are among the important inputs of the software, the 20-years average of the wind speed during different months in the stations under investigation, obtained from NASA's website (Vahdatpour et al., 2018), are provided in Table 3 Appendix. Because of the fact that the altitude of the station under study from the sea level, affects the air density and, therefore, the production capability of the wind turbine, the altitudes of different stations are also provided in Table 3 Appendix.

Since the purpose of the present study is to provide electricity for a household, daily and seasonal electricity profiles are illustrated in Figs. 6-a and b, respectively. Moreover, since the HOMER software performs technical-economic analysis, Table 2 provides the price list for the equipment used and their technical information along with schematics of the system under study.

As seen from Table 2, tree-shaped wind turbines produce AC power, and vertical axis wind turbines produce DC power. In other words, an electrical converter is needed when using vertical axis wind turbines in order to convert the generated DC power into the required AC power.

In the analysis of the performance of wind turbines, the production power curve in terms of the wind speed in the altitude of the hub is of great importance, because the speed of the wind passing through the wind turbine is always variable. Table 3 shows the power curves of the wind turbines used, as well as their shapes and capacities. It is seen from the power curves that in addition to having different capacities, the cut-in and cut-off speeds of the turbines are also different, which results in different performances in wind turbines.

3. Methodology

3.1. SWARA method

The step-wise assessment ratio analysis (SWARA) method, is of the newest methods of weighting the indices, which was proposed by Keršuliene et al. in 2010. The most important advantage of this method compared to other similar methods is its ability to evaluate the accuracy of experts (Keršuliene et al., 2010), and that the experts could consult with each other during the process, which would make the results more accurate (Dehnavi et al., 2015). In addition, this method is simple and understandable and contains a smaller number of pairwise comparisons (Stanujkic et al., 2015). The weighting steps based on the SWARA method are organized as follows (Keršuliene et al., 2010):

- Step 1: arranging the indices;
 - Step 2: determining the relative importance of every index (S_j) used in this study;
 - Step 3: calculating the K_j coefficient;
- K_j is a function of the relative importance of every index used in this study, which can be calculated as follows:

$$K_j = S_j + 1 \quad (1)$$

- Step 4: calculating the initial weight of every used index;
- The initial weight of each index is calculated using Eq. (2). It should be noted that in this equation, the weight of the first index, the most important index, is assumed to be one.

$$q_j = \frac{q_{j-1}}{K_j} \quad (2)$$

- Step 5: calculating the final normal weight;
- In the final step of the SWARA method, the final weights of the indices used in this research (called normalized weights) are calculated as the following equation:

$$w_j = \frac{q_j}{\sum q_j} \quad (3)$$

3.2. WSM method

The weighted sum method (WSM) method which is also known as the simple additive weighted (SAW) method, is one of the simplest and most common MCDM methods and involves the summation of the values of the criteria for each option and the use of the weights of the criteria separately (Fishburn, 1967). In general, WSM deals with profit criteria. It means that the cost criteria are transformed into profit criteria. After the transformation, the lowest cost criterion is transformed into the highest criterion, and the highest cost is transformed into the lowest criterion. The decision matrix is normalized linearly. Then the value of each criterion is multiplied by its weight. Consequently, the summation of the values is calculated for each

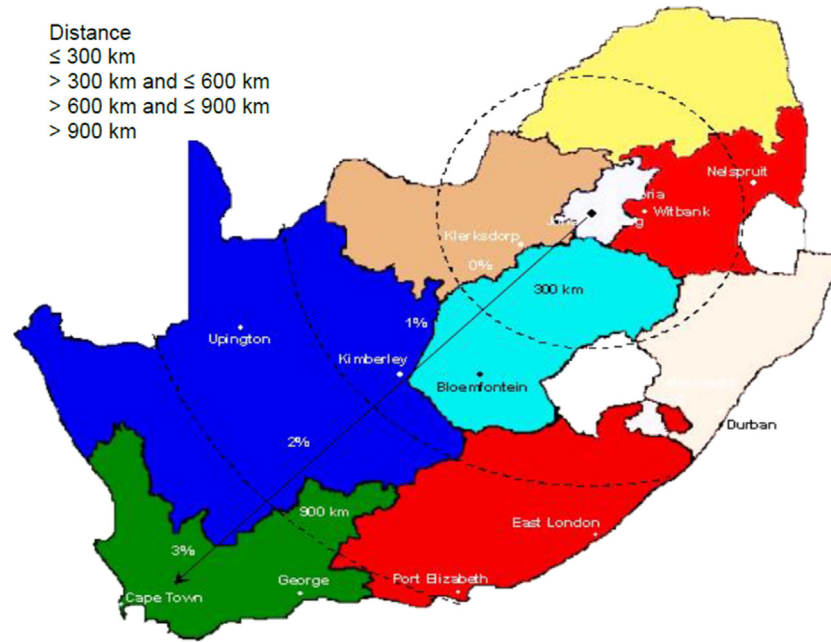


Fig. 4. Transmission zones for loads (Anon., 2020c).

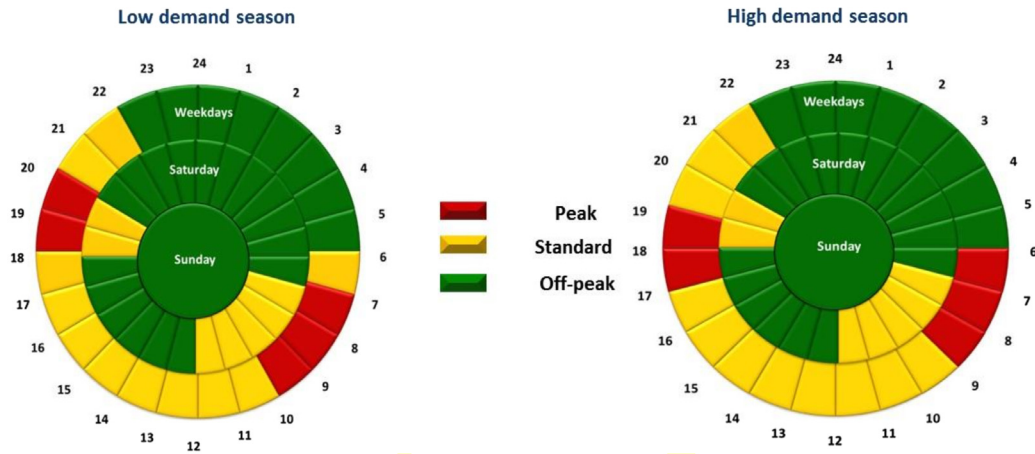


Fig. 5. Time-of-use periods (Anon., 2020c).

option, and the option with the highest value is chosen as the best solution (Fishburn, 1967):

$$A_{WSM}^* = \max_j \sum_{i=1}^M w_i a_{ij} \quad (4)$$

In the $M \times N$ matrix, the data inputs a_{ij} denote the value of the j th option in terms of the i th decision criterion. A^* denotes the score of the WSM optimal option, and w_i is the standard weight of the i th decision. In this method, the option that is selected has a higher value of scale-less summation of its weight values, compared to the other options.

3.3. ARAS method

Additive ratio assessment (ARAS) is a novel method for ranking the options based on the degree of optimality. It is derived by dividing the summation of weighted and normalized values of the criteria for each option, which is an indication of the status of each option, by the summation of the weighted and normalized values for the most appropriate option. In this method, the

degree of optimality is calculated (Zavadskas and Turskis, 2010; Zavadskas et al., 2010). The steps of the ARAS method are as follows (Zavadskas and Turskis, 2010):

The decision matrix is formed in the first step. It is an $m \times n$ matrix where m denotes the number of options (rows) while n is the number of criteria (columns).

$$X = \begin{bmatrix} x_{01} & \dots & x_{0j} & \dots & x_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mj} & \dots & x_{mn} \end{bmatrix} \quad i = \overline{0, m}; j = \overline{1, n} \quad (5)$$

where m is the number of options, and n is the number of criteria. x_{ij} denotes the performance of the i th option in the j th criterion. x_{0j} represents the optimal value for the j th criterion. If the optimal value of the j th variable is indefinite, we set a value for it as

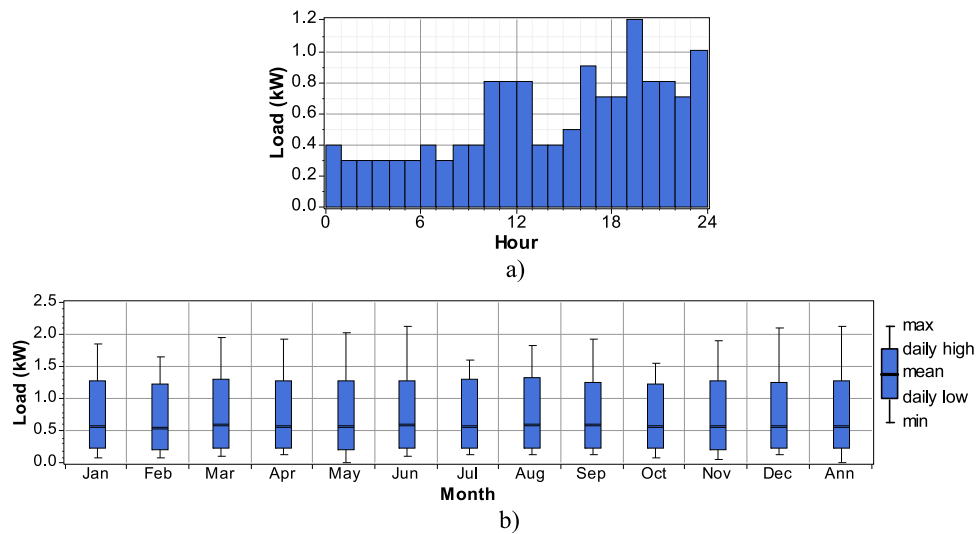


Fig. 6. Electricity profile (a) Daily (b) Seasonal.

Table 2
System information.

Equipment	Cost (\$)			Size (kW)	Other information	Schematic
	Capital	Replacement	O&M			
Converter (Jahangiri et al., 2018, 2019)	800	700	100	1	Lifetime: 15 years Efficiency: 90%	
EOLO vertical axis wind turbine (VAWT) (Anon., 2017b)	5269	5269	130	3	Lifetime: 20 years Hub height: 10 m	
Spiral vertical axis wind turbine (VAWT) (Anon., 2017c)	1900	1900	48	1	Lifetime: 20 years Hub height: 10 m	
Turby vertical axis wind turbine (VAWT) (Syngellakis et al., 2005)	19243	19243	480	2.5	Lifetime: 20 years Hub height: 10 m	
WRE vertical axis wind turbine (VAWT) (Anon., 2017a)	13635	13635	340	3	Lifetime: 20 years Hub height: 10 m	
Tree shape wind turbine (TSWT) vertical axis wind turbine (VAWT) (Mostafaeipour et al., 2019)	35000	35000	200	11.7	Lifetime: 19 years Hub height: 6.8 m	

follows:

$$\begin{aligned}
 x_{0j} &= \max_i x_{ij}, & \text{if } \max_i x_{ij} & \text{ is preferable} \\
 x_{0j} &= \min_i x_{ij}^*, & \text{if } \min_i x_{ij}^* & \text{ is preferable}
 \end{aligned} \quad (6)$$

The evaluation value for the options in the criteria (x_{ij}) and the weight of each criterion (w_j), are usually used by the decision-makers as the inputs of the decision matrix method (DMM). It should be noted in the first step that the criteria have different dimensions. In order to make the comparison of the criteria possible and to avoid the possible difficulties due to the differences in the dimensions of the criteria, we must first make the weighted values dimensionless. To this end, we divide the values to the optimal value obtained above. There exist different methods for making the values dimensionless, one of which is described here.

Using the normalization method, the elements of the initial decision matrix are transformed into values in the range of (0, 1) or (0, ∞).

In the second step, the initial input values are normalized for all of the criteria and written as $\bar{x}_{ij}$ which are the elements of the

matrix $\bar{X}$, defined as follows:

$$\bar{X} = \begin{bmatrix} \bar{x}_{01} & \dots & \bar{x}_{0j} & \dots & \bar{x}_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{x}_{i1} & \dots & \bar{x}_{ij} & \dots & \bar{x}_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{x}_{m1} & \dots & \bar{x}_{mj} & \dots & \bar{x}_{mn} \end{bmatrix} \quad i = \overline{0, m}; j = \overline{1, n} \quad (7)$$

The normalization for the benefit type criteria is performed as in the following:

$$\bar{x}_{ij} = \frac{x_{ij}}{\sum_{i=0}^m x_{ij}} \quad (8)$$

The normalization for the cost type criteria is performed as follows:

$$x_{ij} = \frac{1}{x_{ij}^*} \quad \bar{x}_{ij} = \frac{x_{ij}}{\sum_{i=0}^m x_{ij}} \quad (9)$$

Once the dimensionless values of the criteria are determined, it would be possible to compare the criteria to each other.

In the third step: the weights are applied to the normalized matrix $\bar{X}$, to obtain the matrix $\hat{X}$. The weight of the j th criterion is

Table 3
Data of VAWTs used.

Turbine type	Power curve	Figure
EOLO 3 kW		
Spiral 1 kW		
Turby 2.5 kW		
WRE 3 kW		
TSWT (Aeroleaf) 11.7 kW		

denoted by w_j . The weights are determined by experts. The given weights must satisfy the following conditions:

$$0 < w_j < 1$$

$$\sum_{j=1}^n w_j = 1 \quad (10)$$

$$\hat{X} = \begin{bmatrix} \hat{x}_{01} & \dots & \hat{x}_{0j} & \dots & \hat{x}_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{x}_{i1} & \dots & \hat{x}_{ij} & \dots & \hat{x}_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{x}_{m1} & \dots & \hat{x}_{mj} & \dots & \hat{x}_{mn} \end{bmatrix} \quad i = \overline{0, m}; j = \overline{1, n} \quad (11)$$

$$\hat{x}_{ij} = \bar{x}_{ij} \times w_j; \quad i = \overline{0, m} \quad (12)$$

where w_j is the weight (importance) of the j th criterion, and $\bar{x}_{ij}$ is the normalized value of the j th criterion. The following equation defines the value of the optimality function:

$$S_i = \sum_{j=1}^n \hat{x}_{ij}; \quad i = \overline{0, m} \quad (13)$$

where S_i denotes the value of the optimality function for the i th option. The most appropriate option is the one that has the maximum optimality function value, and the worst case is the minimum optimality function value. The priority of the options is defined by the value of their corresponding S_i .

The alternative utility of every option is calculated from its comparison to the best value of optimality denoted by S_0 . The

utility degree equation, shown by K_j is described below for A_i .

$$K_i = \frac{S_i}{S_0}; \quad i = \overline{0, m} \quad (14)$$

The options are ranked based on their corresponding K_i values.

3.4. WASPAS method

The weighted aggregated sum product assessment (WASPAS) method is one of the novel decision-making techniques introduced by [Zavadskas et al. \(2012\)](#) in 2012. This method was derived based on a combination of the WSM (weighted sum method) and WPM (weighted product model) methods and can be very useful in complex decision-making problems; moreover, the results from this method have high accuracy.

First, the decision matrix is formed. The dimension of this matrix is $m \times n$, where m denotes the number of options (rows), and n is the number of criteria (columns).

$$X = \begin{bmatrix} x_{01} & \dots & x_{0j} & \dots & x_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{i1} & \dots & x_{ij} & \dots & x_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ x_{m1} & \dots & x_{mj} & \dots & x_{mn} \end{bmatrix} \quad i = \overline{0, m}; j = \overline{1, n} \quad (15)$$

where m is the number of options, and n is the number of criteria. x_{ij} denotes the performance of the i th option in the j th criterion. x_{0j} is the optimal value of the j th criterion. If the optimal value of the j th variable is indefinite, a value is determined for it in the following routine.

$$\begin{aligned} x_{0j} &= \max_i x_{ij}, & \text{if } \max_i x_{ij} & \text{is preferable} \\ x_{0j} &= \min_i x_{ij}^*, & \text{if } \min_i x_{ij}^* & \text{is preferable} \end{aligned} \quad (16)$$

In the first step, it should be noted that the criteria are of different dimensions.

In the second step, the initial input values are normalized for all criteria and written in the form of $\bar{x}_{ij}$, which are the elements of the matrix $\bar{X}$ defined as follows:

$$\bar{X} = \begin{bmatrix} \bar{x}_{01} & \dots & \bar{x}_{0j} & \dots & \bar{x}_{0n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{x}_{i1} & \dots & \bar{x}_{ij} & \dots & \bar{x}_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{x}_{m1} & \dots & \bar{x}_{mj} & \dots & \bar{x}_{mn} \end{bmatrix} \quad i = \overline{0, m}; j = \overline{1, n} \quad (17)$$

The normalization for the benefit type criteria is given by:

$$\bar{x}_{ij} = \frac{x_{ij}}{\max_i x_{ij}} \quad (18)$$

The normalization for the cost type criteria is calculated as:

$$x_{ij} = \frac{\min_i x_{ij}}{x_{ij}^*} \quad (19)$$

Step 3-a: calculation of the normalized decision matrix for the WSM model

$$\hat{X}_q = \begin{bmatrix} \hat{x}_{11} & \dots & \hat{x}_{1j} & \dots & \hat{x}_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{x}_{i1} & \dots & \hat{x}_{ij} & \dots & \hat{x}_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \hat{x}_{m1} & \dots & \hat{x}_{mj} & \dots & \hat{x}_{mn} \end{bmatrix}; \quad (20)$$

$$\hat{x}_{ij} = \bar{x}_{ij} w_j, \quad i = \overline{1, m}; j = \overline{1, n}$$

Step 3-b: calculation of the normalized decision matrix for the WPM model:

$$\bar{\bar{X}}_p = \begin{bmatrix} \bar{\bar{x}}_{11} & \dots & \bar{\bar{x}}_{1j} & \dots & \bar{\bar{x}}_{1n} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{\bar{x}}_{i1} & \dots & \bar{\bar{x}}_{ij} & \dots & \bar{\bar{x}}_{in} \\ \vdots & \ddots & \vdots & \ddots & \vdots \\ \bar{\bar{x}}_{m1} & \dots & \bar{\bar{x}}_{mj} & \dots & \bar{\bar{x}}_{mn} \end{bmatrix}; \quad (21)$$

$$\bar{\bar{x}}_{ij} = \bar{x}_{ij}^{w_j}, \quad i = \overline{1, m}; j = \overline{1, n}$$

Step 4: calculation of the value of the optimal function:

(a) From the WSM model, the following holds for every option:

$$Q_i = \sum_{j=1}^n \hat{x}_{ij}, \quad i = \overline{1, m} \quad (22)$$

(b) From the WPM model, the following holds for every option:

$$P_i = \sum_{j=1}^n \bar{\bar{x}}_{ij}, \quad i = \overline{1, m} \quad (23)$$

Step 5: final calculations to evaluate and prioritize the options according to:

$$WPS_i = 0.5 \sum_{j=1}^m Q_i + 0.5 \sum_{j=1}^m P_i \quad (24)$$

In the WASPAS method, the options are ranked based on the values of WPS_i .

3.5. Modeling in HOMER software

Economic output is one of the essential outputs that should be noted in the system design; thus, the HOMER provides a list of the systems classified based on the total NPC of the grid ([Singh and Baredar, 2016](#)). The total NPC represents all the costs imposed on the system throughout the lifetime of the project (including installation, replacement, fuel, purchase of electricity from the grid and the penalties due to the emission of pollutants) minus the revenues of the system (including the revenues from selling electricity to the grid or the revenues from selling the scrap equipment). The total NPC is calculated as follows ([Naderi et al., 2017](#)):

$$NPC = \frac{C_{ann, total}}{CRF(i, R_{proj})} \quad (25)$$

where $C_{ann, total}$ is the total annual cost, CRF is the return on investment factor, i denotes the actual interest rate, and R_{proj} is the lifetime of the project. All the costs and revenues are assessed at a constant interest rate throughout the year. In this assessment, in order to involve the inflation in the calculations, the actual interest rate resulting from the inflation is calculated, and the impact of changes in the interest rate is applied on the final net cost.

The return on the capital factor, which represents the return on the investment over the course of N years, is calculated from the following equation ([Suresh et al., 2020](#)):

$$CRF = \frac{i(1+i)^N}{(1+i)^N - 1} \quad (26)$$

The software is capable of calculating the actual annual interest rate from the following equation ([Zavadskas et al., 2012](#); [Singh and Baredar, 2016](#)):

$$i = \frac{i' - f}{1 + f} \quad (27)$$

Moreover, the cost per kWh of energy during the useful lifetime of the project is calculated by the software using the following equation (Kasaeian et al., 2020):

$$\text{COE} = \frac{C_{\text{ann, total}}}{E_{\text{LoadServed}}} \quad (28)$$

In the equation above, $E_{\text{loadServed}}$ denotes the actual electric load from the hybrid system with the unit of kWh/y and in terms of \$.

The power curve usually determines the performance of the wind turbine under standard temperature and pressure conditions. To tune the real conditions, HOMER multiplies the predicted power from the power curve to the air density ratio, and the output power of the wind turbine is calculated as follows (Anon., 2016).

$$P_{\text{WTG}} = \frac{\rho}{\rho_0} \times P_{\text{WTG, STP}} \quad (29)$$

where ρ is the real air density, ρ_0 is the air density under the standard temperature and pressure conditions, and $P_{\text{WTG, STP}}$ denotes the output power of the wind turbine derived from the power curve. The value of ρ in Eq. (29) is calculated from the following:

$$\frac{\rho}{\rho_0} = \left(1 - \frac{Bz}{T_0}\right)^{\frac{g}{RB}} \times \left(\frac{T_0}{T_0 - Bz}\right) \quad (30)$$

where z denotes the height in terms of m, B is the rate of decrease equal to 0.0065 K/m, g is the gravitational acceleration that is equal to 9.81 m/s², and T_0 denotes the standard temperature set equal to 288.16 K.

4. Data analysis

4.1. Analysis of the results from HOMER

The simulation results using the HOMER software for different wind turbines are presented in Tables 4–8 Appendix in the form of decision matrices for each wind turbine.

The results show for the TSWT turbine, the minimum and maximum price per kWh of produced energy, with the values of 0.106 \$ and 1.465 \$, belong to Port Elizabeth, and Bloemfontein, respectively. The reason for this is the power generated by the wind turbine, because these two stations, producing 7380 kWh, and 1555 kWh annually, have generated the highest and lowest amounts of electricity, respectively. As seen in Table 4 Appendix, the production of CO₂ pollutant is negative in the three stations of East London, Port Elizabeth, and Cape Town, which means that the emission of pollutants has been prevented. Additionally, it is seen from the results that the revenue of the wind system for each city, which is presented in comparison to only using the national electricity grid, is negative for the four stations that have generated the lowest amount of wind electricity. This means that the use of TSWT wind turbines is not an economic priority in these stations, and 100% usage of the power grid would be more cost-efficient.

It is seen from the results of Table 5 Appendix, which is related to the EOLO vertical axis wind turbine, that the average output power has increased compared to the TSWT turbine, which is due to the higher generation of wind electricity by EOLO compared to TSWT. For the same reason, the emission of produced pollutants is also lower for EOLO compared to TSWT. Moreover, given that in some stations, the total price of the electricity sold to the national electricity grid is higher than the price of the electricity bought from the grid, the LCOE value in the four stations of Port Elizabeth, East London, Durban, and Cape Town is negative. Consequently, the amount of CO₂ pollutant emission would also be negative in

these four stations, indicating that no pollutant is emitted. The lowest amount of LCOE, which results in annual revenue of 9597 \$, with a value of −0.363 \$/kWh, belongs to the station of Port Elizabeth.

The results in Table 6 Appendix, corresponding to the Spiral vertical axis wind turbine, show that pollutant emission exists in all stations because of using the national electricity grid and the Bloemfontein station with an annual 2772 kg emission of CO₂ pollutant has the worst environmental condition among the stations under study. The reason for the higher emission of pollutants in spiral wind turbine compared to TSWT and EOLO wind turbines is that it produces less wind electricity compared to them, due to the shorter hours of wind availability for this turbine. Similar to the previous two wind turbines, the Port Elizabeth station with an LOE equal to 0.582 \$/kWh has the lowest cost among the stations under investigation for the spiral vertical axis wind turbine. The lowest annual pollutant generation and the highest annual revenue, belong to the Port Elizabeth station, with values of 1513 kg and 2678 \$, respectively.

The important issue seen from the results in Table 7 Appendix, which corresponds to the Turby vertical axis wind turbine, is that due to the negative income of all stations, the use of wind turbine is not the first economic option in any of the stations and the use of the power grid alone would be more cost-efficient. Similar to the spiral wind turbine, the pollutant emission also exists in all stations for the Turby wind turbine, which is due to the very low amount of the generated wind electricity. The highest hours of wind turbine performance, the highest average output power, the lowest amount of generated pollutant, and the lowest LCOE, all correspond to the Port Elizabeth station, with the values of 6767 h, 0.24 kW, 2022 kg/year, and 1.127 \$/kWh, respectively.

The results for the WRE vertical axis wind turbine are presented in Table 8 Appendix. It is seen from these results that CO₂ pollutant emission exists in all stations, which is due to the poor performance of the wind turbine in providing the total amount of electricity required. The LCOE range for the WRE vertical axis wind turbine varies from 1.02 \$/kWh in Port Elizabeth to 1.426 \$/kWh in Bloemfontein. Another point that can be seen from the results is that only in two stations, Port Elizabeth and Cape Town, the use of wind turbines is the first economic priority, and the use of the national electricity grid is in the second rank of economic priority.

In general, it can be stated from the results in Tables 4–8 Appendix, that the Port Elizabeth station has the lowest LCOE equal to −0.363 \$/kWh, when using EOLO wind turbine and the Bloemfontein station has the highest LCOE with the value of 1.601 \$/kWh, when using the Turby wind turbine.

4.2. Weighting the indices with SWARA method

In the current work, a decision-making six-member team, including academic experts and field practitioners who possess professional knowledge and expertise in the field of renewable energy, was used for weighting the indices. Based on the SWARA method, it was asked by the experts to arrange the criteria in descending order of importance. Then the indices for the technical and economic specifications were weighted, as shown in Table 9 Appendix.

Then, each of the sub-indices corresponding to technical and economic specifications was weighted according to the SWARA technique, and their results are represented in Tables 10 and 11 Appendix.

Consequently, the weight of each index is multiplied by the weights of its sub-indices to calculate the final weight of each sub-index. These calculations are shown in Table 12 Appendix.

Weighting results showed that wind penetration, total production, and capital cost sub-indices with weights equal to 0.106,

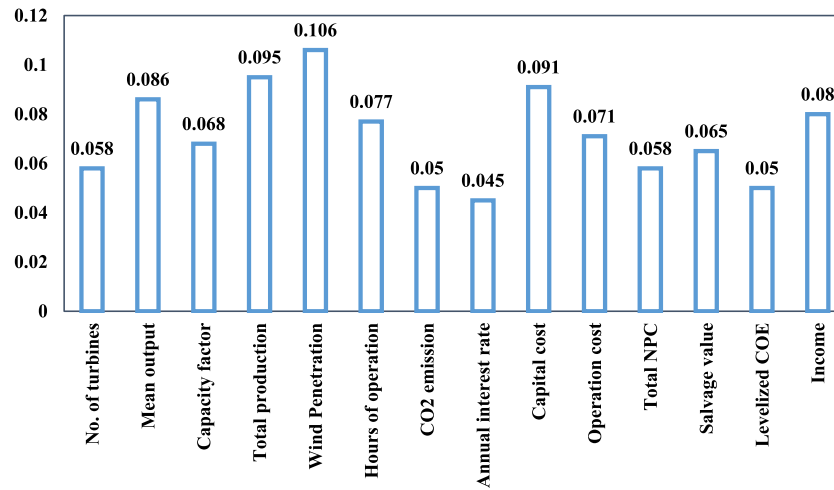


Fig. 7. Comparison between the final weights of the sub-indices.

Table 4
Ranking of cities by the WSM method.

City	A _{WSM}					Rank
	TSWT	EOLO	Spiral	Turby	WRE	
Klerksdorp	0.539	0.519	0.708	0.638	0.695	5
Witbank	0.517	0.486	0.668	0.596	0.662	7
Bloemfontein	0.503	0.467	0.631	0.565	0.637	8
Durban	0.587	0.584	0.780	0.707	0.758	4
East London	0.682	0.654	0.887	0.857	0.881	3
Port Elizabeth	0.920	0.972	0.963	0.967	0.973	1
Cape Town	0.769	0.696	0.937	0.935	0.951	2
George	0.532	0.516	0.693	0.611	0.679	6

0.095, 0.091, respectively, were identified as the most important sub-indices. Fig. 7 illustrates the comparison between the final weights of the sub-indices in a diagram.

4.3. Ranking the South African cities

The cities under study are ranked using WSM, ARAS, and WASPAS techniques, and the final rank of each city is determined based on the average rank of the cities. In order to rank the cities, decision matrices were first developed, which are presented in Tables 4–8 Appendix for different wind turbines.

The decision matrix in the WSM method is normalized linearly. The value for each index is then multiplied by its corresponding weight. Then the sum values are calculated for every option, and the option with the highest value is chosen as the most suitable solution. The ranking of the cities based on the WSM method is presented in Table 4.

In the ARAS technique, first, the decision matrix is molded, and then it is normalized based on whether the indices are benefit-type or cost-type. Moreover, the normalized matrix is weighted and the value of an optimal function (S_i) is calculated. Also, the most appropriate option is the one that has the highest value for the optimal function, and the worst case is the lowest value of the optimal function. The priority of the options is defined based on the S_i value. The alternative utility for every option is determined from its comparison with the best value, named S_0 . The equation of alternative utility, shown by K_i , is calculated for every option. In conclusion, the options are ranked based on their K_i values. The results for the ranking of the cities using the ARAS method are presented in Table 5.

According to the WASPAS method, first, the decision matrix is normalized, and then the normalized decision matrix is calculated

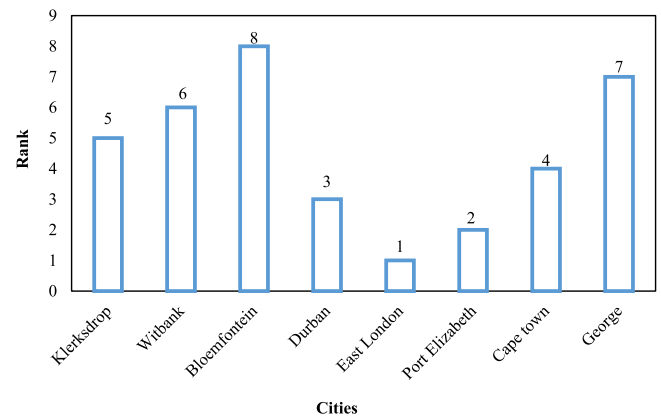


Fig. 8. Ranking results of cities under study.

for both WSM and WPM methods. Then the values Q_i and P_i are calculated. After that, the WPS_i values are calculated for every city. The rankings are performed based on the WPS_i values. The results of ranking based on WASPAS method are presented in Table 6.

Now for the final ranking of the cities, the average of the results from WSM, ARAS and WASPAS techniques is taken, the results of which are presented in Table 7. The results identified Port Elizabeth, East London, and Durban as first, second, and third suitable cities for using different types of wind turbines. Moreover, Bloemfontein was identified as the worst city for this purpose. The ranking results of cities in Fig. 8 are presented in the form of a comparison chart.

5. Conclusion

South Africa faces serious challenges in the energy sector in spite of having vast natural resources, that if properly addressed, not only resolve its own demands, but also would prevent some global phenomena such as desertification, destruction of the environment and emission of greenhouse gases. To address the importance of the aforementioned challenges, and since no technical-economic-environmental study has been conducted on the potential of using the new generation of wind turbines at the residential building scale in South Africa, the present study investigated these issues using HOMER software on the 20-year average wind data. Then used the SWARA weighting method

Table 5
Ranking of the cities based on the ARAS method.

City	TSWT		EOLO		Spiral		Turby		WRE		Rank
	S_i	K_i	S_i	K_i	S_i	K_i	S_i	K_i	S_i	K_i	
Klerksdorp	0.065	0.288	0.080	0.381	0.095	0.669	0.085	0.543	0.091	0.623	5
Witbank	0.059	0.264	0.077	0.365	0.088	0.623	0.077	0.491	0.085	0.581	7
Bloemfontein	0.055	0.245	0.076	0.363	0.082	0.580	0.071	0.454	0.080	0.548	8
Durban	0.077	0.342	0.087	0.416	0.107	0.754	0.099	0.634	0.103	0.704	4
East London	0.103	0.458	0.096	0.456	0.124	0.878	0.130	0.830	0.126	0.861	3
Port Elizabeth	0.224	1.000	0.190	0.907	0.137	0.964	0.153	0.974	0.143	0.977	1
Cape Town	0.130	0.577	0.104	0.495	0.133	0.936	0.147	0.934	0.139	0.950	2
George	0.063	0.281	0.080	0.380	0.093	0.653	0.080	0.511	0.088	0.603	6

Table 6
Ranking of the cities according to WASPAS.

City	TSWT			EOLO			Spiral			Turby			WRE			Rank
	Q_i	P_i	WPS_i	Q_i	P_i	WPS_i	Q_i	P_i	WPS_i	Q_i	P_i	WPS_i	Q_i	P_i	WPS_i	
Klerksdorp	12.843	0.539	6.691	12.850	0.572	6.711	13.610	0.708	7.159	13.391	0.638	7.015	13.539	0.695	7.117	3
Witbank	12.751	0.517	6.634	12.767	0.549	6.658	13.520	0.668	7.094	13.211	0.596	6.904	13.434	0.662	7.048	4
Bloemfontein	12.667	0.503	6.585	12.654	0.546	6.600	13.414	0.631	7.022	13.031	0.565	6.798	13.329	0.637	6.983	5
Durban	12.997	0.587	6.792	12.988	0.620	6.804	13.737	0.780	7.259	13.589	0.707	7.148	13.683	0.758	7.221	2
East London	13.237	0.682	6.960	13.173	0.660	6.917	13.875	0.887	7.381	13.840	0.857	7.348	13.868	0.881	7.375	1
Port Elizabeth	10.000	0.747	5.373	9.977	0.726	5.352	10.000	0.747	5.373	10.000	0.747	5.373	10.000	0.747	5.373	6
Cape Town	9.647	0.627	5.137	9.642	0.599	5.120	9.978	0.725	5.352	9.969	0.716	5.342	9.979	0.726	5.352	7
George	9.048	0.387	4.718	9.231	0.449	4.840	9.641	0.491	5.066	9.332	0.388	4.860	9.530	0.457	4.993	8

Table 7
The final ranking of the cities.

City	WSM	ARAS	WASPAS	Average ranking	Final rank
Klerksdorp	5	5	3	4.333	5
Witbank	7	7	4	6.000	6
Bloemfontein	8	8	5	7.000	8
Durban	4	4	2	3.333	3
East London	3	3	1	2.333	1
Port Elizabeth	1	1	6	2.667	2
Cape Town	2	2	7	3.667	4
George	6	6	8	6.667	7

to weigh the technical and economic indices, and identifies the most important sub-indices. Finally, the stations were ranked using WSM, ARAS, and WASPAS techniques, and the most suitable and unsuitable stations were identified. The main conclusions include:

- The Port Elizabeth station with -0.363 \$/kWh has the lowest value of LCOE in comparison to other stations when using EOLO wind turbine.
- The Bloemfontein station with 1.601 \$/kWh has the highest value of LCOE in comparison to other stations when using Turby wind turbine.
- The wind penetration with final weight of 0.106 was identified as the most important sub-index.
- The East London station was identified as the most appropriate station for using a variety of wind turbines.
- The Bloemfontein station was identified as the most unsuitable station for using a variety of wind turbines.

CRedit authorship contribution statement

Ali Mostafaeipour: Conceptualization, Writing - original draft. **Mehdi Jahangiri:** Software, Methodology. **Ahmad Haghani:** Methodology, Writing. **Seyyed Jalaladdin Hosseini Dehshiri:** Writing - review & editing, Validation. **Seyyed Shahabaddin Hosseini Dehshiri:** Writing - review & editing, Software. **Ahmad Sedaghat:** Supervision, Data curation. **Hamed Saghaei:** Data curation. **Esther T. Akinlabi:** Data curation, Software. **Sam M. Sichilalu:** Data curation, Software, Validation. **Shahariar Chowdhury:** Revise, Funding acquisition, Resources. **Kuaanan Techato:** Supervision, Resources, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

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